

MLA0303-Reinforcement Learning

CO-3 AT-3

Data Interpretation

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1. Increasing Reward from 10 to 45

1. Data Observation

The reward values increase gradually from 10 to 45 as the number of training episodes increases. This is a clear positive trend in the learning curve. The agent is receiving better cumulative feedback as training progresses.

2. RL Concept and Interpretation

In Reinforcement Learning, reward represents the quality of the outcome produced by the agent's actions. During early episodes, the agent has limited knowledge and performs more exploratory or incorrect actions, so the reward is low. With repeated interaction, the agent updates its policy or value estimates and increasingly selects actions that produce better future outcomes.

3. Analysis and Justification

The reward increases by 35 points, from 10 to 45. This represents a substantial improvement in learning performance. A gradually increasing reward generally indicates that the policy is becoming more effective. If the reward curve later becomes stable around a high value, it provides stronger evidence that the agent is approaching convergence. However, an increasing reward alone cannot prove that the policy is globally optimal. Additional evidence such as success rate, moving-average reward, reward variance and evaluation on unseen episodes should be considered.

4. Conclusion

Therefore, the increase from 10 to 45 strongly indicates successful learning and policy improvement. The agent is gradually discovering better actions and achieving higher cumulative returns. If the reward remains high and stable in later episodes, the result would provide strong evidence of convergence toward an effective policy.

2. Epsilon-Greedy: $\epsilon = 0.1, 0.3$ and 0.5

1. Data Observation

The cumulative reward is highest when $\epsilon = 0.1$ and lowest when $\epsilon = 0.5$. Therefore, the observed performance decreases as the exploration probability increases from 0.1 to 0.5.

2. RL Concept and Interpretation

In an ϵ -greedy strategy, ϵ represents the probability of exploration. With $\epsilon = 0.1$, the agent explores approximately 10% of the time and exploits its currently best-known action about 90% of the time. With $\epsilon = 0.5$, exploration occurs approximately half of the time, so the agent frequently selects actions that may have lower estimated value.

3. Analysis and Justification

The result shows that exploitation is more beneficial under the tested conditions. At $\epsilon = 0.1$, the agent retains a small amount of exploration while mainly using the learned policy. At $\epsilon = 0.3$, more decisions are spent testing alternatives, reducing immediate cumulative reward. At $\epsilon = 0.5$, excessive random exploration causes the largest reduction in reward. However, $\epsilon = 0.1$ is not universally optimal; early training may require greater exploration. A practical approach is to start with a larger ϵ and gradually reduce it.

4. Conclusion

Among the tested values, $\epsilon = 0.1$ provides the best performance because it achieves the highest cumulative reward. The result demonstrates the exploration–exploitation trade-off: controlled exploration is necessary for learning, but stronger exploitation becomes more effective after useful action values have been discovered.

3. Q-Learning vs SARSA

1. Data Observation

Q-Learning consistently produces slightly higher average rewards than SARSA at 100, 200 and 300 episodes. The difference remains in favor of Q-Learning at every reported training point.

2. RL Concept and Interpretation

Q-Learning is an off-policy algorithm that updates toward the maximum estimated value of the next state. SARSA is an on-policy algorithm that updates using the action actually selected by the current behavior policy. Because SARSA accounts for the consequences of the exploratory action that is actually taken, it can learn a more conservative policy when exploration is present.

3. Analysis and Justification

The consistently higher rewards indicate that Q-Learning performs better under the stated experimental conditions. If Q-Learning reaches a strong reward level earlier and maintains that advantage at 100, 200 and 300 episodes, it can reasonably be considered to have faster observed convergence. Strictly, convergence speed should be determined from reward curves, variance and the number of episodes required to reach a defined performance threshold, rather than final reward alone.

4. Conclusion

Based on the given data, Q-Learning is the better-performing algorithm and shows evidence of faster convergence than SARSA in this experiment. The conclusion is specific to the environment, reward structure, exploration strategy and hyperparameters used.

4. Discount Factor: $\gamma = 0.2, 0.5$ and 0.9

1. Data Observation

The average reward increases as the discount factor changes from 0.2 to 0.5 and then to 0.9. Therefore, the highest tested discount factor produces the highest learning performance.

2. RL Concept and Interpretation

The discount factor γ controls how strongly future rewards influence current decisions. A low value such as 0.2 makes the agent focus mainly on immediate rewards. A moderate value such as 0.5 gives more importance to future outcomes. A high value such as 0.9 encourages long-term planning because future rewards retain greater weight.

3. Analysis and Justification

The observed increase indicates that the environment contains important delayed consequences. An agent with $\gamma = 0.9$ can select actions that may provide smaller immediate rewards but lead to better future states and larger cumulative returns. Therefore, $\gamma = 0.9$ is the most suitable value among the tested choices. Nevertheless, a very high discount factor can sometimes make learning more sensitive to uncertain distant outcomes, so stability and convergence should also be checked.

4. Conclusion

$\gamma = 0.9$ provides the best observed learning performance because it gives greater importance to long-term rewards. The trend demonstrates that effective decision-making in this environment requires the agent to consider future consequences rather than optimizing only immediate rewards.

5. Success Rate: 45% to 92%

1. Data Observation

The success rate increases from 45% at 50 episodes to 92% at 200 episodes. This is an improvement of 47 percentage points, showing a substantial increase in the agent's ability to complete the task successfully.

2. RL Concept and Interpretation

Success rate measures how often the learned policy achieves the required objective. The increase indicates that additional training allows the agent to improve its action selection. Early episodes contain more exploration and mistakes, while later episodes contain more actions based on learned experience.

3. Analysis and Justification

The relative increase compared with the initial 45% is approximately 104.4%. A 92% success rate means the agent succeeds in most tested episodes and provides strong evidence of policy improvement. However, 92% alone does not prove global optimality. To establish optimal or near-optimal behavior, the agent should be tested on unseen states or random seeds, compared with a known baseline or theoretical optimum, and checked for continued stability after 200 episodes.

4. Conclusion

The increase from 45% to 92% strongly demonstrates learning and major policy improvement. The agent has learned a highly effective policy under the tested conditions. It can be considered converged to a strong or near-optimal policy only if the 92% success rate remains stable and is close to the known best achievable performance.

6. CartPole Learning Rate: 0.01, 0.05 and 0.1

1. Data Observation

The agent achieves higher training stability when the learning rate is 0.05 compared with 0.01 and 0.1. Therefore, 0.05 provides the most stable learning behavior among the tested values.

2. RL Concept and Interpretation

The learning rate determines the size of each parameter or value update. A very small learning rate such as 0.01 produces small changes, which can make learning stable but slow. A large learning rate such as 0.1 produces aggressive updates that can cause oscillation, overshooting or unstable reward curves. A value of 0.05 provides an intermediate update size.

3. Analysis and Justification

The result indicates that 0.05 gives a good balance between learning speed and stability. If 0.01 takes too long to reach a strong reward and 0.1 fluctuates considerably, 0.05 is the most practical choice. The decision should be supported using convergence episodes, moving-average reward, reward variance and final CartPole performance.

4. Conclusion

The learning rate of 0.05 is the best choice among the tested values because it provides the observed balance between sufficiently fast learning and stable convergence. This demonstrates the importance of hyperparameter tuning in achieving reliable RL performance.

7. Exploration Strategies: ϵ -Greedy, Softmax and Random

1. Data Observation

The cumulative reward is highest when ϵ -greedy exploration is used. Therefore, ϵ -greedy provides the best observed performance among the three tested strategies.

2. RL Concept and Interpretation

ϵ -greedy normally exploits the best-known action with high probability and explores randomly with probability ϵ . Softmax exploration assigns probabilities according to estimated action values, while random exploration chooses actions without using learned value information. Random exploration therefore wastes more decisions on potentially poor actions.

3. Analysis and Justification

The higher reward achieved by ϵ -greedy suggests that the environment benefits from controlled exploration combined with strong exploitation. Once useful Q-values have been learned, selecting the best action most of the time produces better cumulative reward. Softmax can also provide informed exploration, but its temperature parameter must be tuned properly. Random exploration lacks the value-guided exploitation provided by both methods.

4. Conclusion

ϵ -greedy achieves better performance in this scenario because it provides a practical balance between exploration and exploitation. The result shows that controlled exploration is more effective than unrestricted random behavior under the given environment and training conditions.

8. Maze Steps Reduce from 50 to 15

1. Data Observation

The agent initially requires 50 steps to reach the maze goal, but after 300 training episodes it requires only 15 steps. The number of steps decreases by 35, which represents a 70% reduction.

2. RL Concept and Interpretation

Fewer steps indicate that the agent has learned a more efficient route. During early training, exploration causes detours, repeated movements and incorrect choices. As the agent gains experience, its policy or Q-values improve and it selects actions that lead more directly toward the goal.

3. Analysis and Justification

The 70% reduction is strong evidence of learning and improved decision-making. If the theoretical shortest path in the maze is 15 steps, the learned policy has effectively reached optimality. If the shortest path is less than 15, the policy is near-optimal rather than fully optimal. Convergence should also be verified by checking whether the step count remains around 15 over later episodes and whether the reward remains stable.

4. Conclusion

The reduction from 50 to 15 steps clearly demonstrates policy improvement and greater navigation efficiency. It provides evidence of convergence toward an optimal or near-optimal policy, depending on the known shortest possible path and the stability of later results.

9. SARSA Discount Factors: $\gamma = 0.4, 0.7$ and 0.95

1. Data Observation

The SARSA agent achieves the highest long-term reward when $\gamma = 0.95$. Therefore, the largest tested discount factor produces the best observed performance.

2. RL Concept and Interpretation

The discount factor controls the balance between immediate and future rewards. $\gamma = 0.4$ strongly emphasizes immediate outcomes, while $\gamma = 0.7$ provides a moderate future perspective. $\gamma = 0.95$ gives high importance to future rewards and encourages the agent to make decisions that improve long-term return.

3. Analysis and Justification

The result indicates that delayed consequences are important in this task. With $\gamma = 0.95$, the SARSA agent is better able to value actions that may have smaller immediate benefits but lead to high future rewards. Therefore, 0.95 is the most effective tested value. Its stability should also be verified because very high discount factors can increase sensitivity to uncertain long-term outcomes.

4. Conclusion

$\gamma = 0.95$ is the most suitable value among the tested choices because it produces the highest long-term reward. The result demonstrates that the SARSA agent benefits from a long-term decision-making horizon in this environment.

10. Reward Stabilization after 500 Episodes

1. Data Observation

The RL agent shows fluctuating rewards during early episodes and then stabilizes after approximately 500 episodes. The transition from high variability to a stable reward pattern indicates that the learning process is becoming more consistent.

2. RL Concept and Interpretation

Early reward fluctuations are expected because the agent is exploring, its policy is changing, and the stochastic environment can produce different outcomes from similar actions. As training continues, the agent gathers more experience and improves its policy or value estimates. Eventually, the selected actions become more consistent, reducing reward variability.

3. Analysis and Justification

Stabilization after 500 episodes provides evidence that the agent has reached a relatively stable learning region. If the stable reward is substantially higher than the early average and the variance remains low, the evidence for policy improvement and robustness is strong. However, stability by itself does not prove optimality because a stable policy can still be suboptimal. Evaluation on independent random seeds and comparison with a baseline are needed.

4. Conclusion

The fluctuating-to-stable reward trend indicates that the agent has progressed from exploration and uncertainty toward a more consistent policy. Stabilization after 500 episodes supports improved learning robustness, particularly if the final reward is high and remains stable during independent evaluation.